《数据库系统原理》实验报告（4）					
题目：SQL 综合练习					
学号	2453359	姓名	李欣洋	日期	4.14

实验环境：MariaDB

实验步骤及结果截图：

1.建立 Library 数据库

```
MariaDB [(none)]> create database Library;
Query OK, 1 row affected (0.007 sec)

MariaDB [(none)]> use Library;
Database changed
MariaDB [Library]>
```

2.建立 Book 表，Student 表，Borrow 表

```
MariaDB [Library]> CREATE TABLE Book (
    ->     Book_no VARCHAR(20) PRIMARY KEY,
    ->     Book_name VARCHAR(100) NOT NULL,
    ->     Author VARCHAR(50),
    ->     Price DECIMAL(10, 2)
    -> );
Query OK, 0 rows affected (0.046 sec)
```

```
MariaDB [Library]> CREATE TABLE Student (
    ->     Student_no VARCHAR(20) PRIMARY KEY,
    ->     Student_name VARCHAR(50) NOT NULL,
    ->     Grade VARCHAR(20)
    -> );
Query OK, 0 rows affected (0.024 sec)
```

```
MariaDB [Library]> CREATE TABLE Borrow (
    ->     Student_no VARCHAR(20),
    ->     Book_no VARCHAR(20),
    ->     Return_date DATE,
    ->     PRIMARY KEY (Student_no, Book_no),
    ->     FOREIGN KEY (Student_no) REFERENCES Student(Student_no),
    ->     FOREIGN KEY (Book_no) REFERENCES Book(Book_no)
    -> );
Query OK, 0 rows affected (0.025 sec)
```

3.插入样例数据

```
MariaDB [Library]> INSERT INTO Book (Book_no, Book_name, Author, Price) VALUES
    -> ('T1001', 'Java 程序设计', '李新珊', 89.5),
    -> ('T1002', '数据库原理及应用', '王敏', 39),
    -> ('T1003', 'Java 高级编程', '陈明海', 63.5),
    -> ('T1004', '专业英语', '张倪宁', 23.1),
    -> ('T1005', 'C++程序设计', '马天颖', 83.2),
    -> ('T1006', '编译原理', '王鑫单', 65);
Query OK, 6 rows affected (0.031 sec)
Records: 6  Duplicates: 0  Warnings: 0

MariaDB [Library]> INSERT INTO Student (Student_no, Student_name, Grade) VALUES
    -> ('K005', '张鑫翁', '大一'),
    -> ('K003', '徐晨皓', '大二'),
    -> ('K002', '王三优', '大三'),
    -> ('K001', '刘孔阴', '大三'),
    -> ('K004', '吴宇涵', '大四');
Query OK, 5 rows affected (0.013 sec)
Records: 5  Duplicates: 0  Warnings: 0

MariaDB [Library]> INSERT INTO Borrow (Student_no, Book_no, Return_date) VALUES
    -> ('K001', 'T1006', '2023-10-9'),
    -> ('K001', 'T1001', '2024-3-1'),
    -> ('K002', 'T1002', '2023-10-9'),
    -> ('K002', 'T1003', '2024-4-5'),
    -> ('K002', 'T1001', '2023-11-3'),
    -> ('K003', 'T1005', '2024-1-4'),
    -> ('K004', 'T1002', '2024-2-5');
Query OK, 7 rows affected (0.016 sec)
Records: 7  Duplicates: 0  Warnings: 0
```

4. 查询书名中包含”程序设计”的图书信息，输出所有信息（包括书名、书号、作者、单价），并按照单价降序排列

```
MariaDB [Library]> SELECT Book_name, Book_no, Author, Price
    -> FROM Book
    -> WHERE Book_name LIKE '%程序设计%'
    -> ORDER BY Price DESC;
+------------------+---------+-----------+-------+
| Book_name        | Book_no | Author    | Price |
+------------------+---------+-----------+-------+
| Java 程序设计    | T1001   | 李新珊    | 89.50 |
| C++程序设计      | T1005   | 马天颖    | 83.20 |
+------------------+---------+-----------+-------+
2 rows in set (0.010 sec)
```

5. 查询借阅了书名为"数据库原理及应用"的学生信息，输出该学生的学号、姓名和年级，并按照学号升序排列

```
MariaDB [Library]> SELECT S.Student_no, S.Student_name, S.Grade
    -> FROM Student S
    -> JOIN Borrow BW ON S.Student_no = BW.Student_no
    -> JOIN Book B ON BW.Book_no = B.Book_no
    -> WHERE B.Book_name = '数据库原理及应用'
    -> ORDER BY S.Student_no ASC;
+------------+--------------+--------+
| Student_no | Student_name | Grade  |
+------------+--------------+--------+
| K002       | 王三优       | 大三   |
| K004       | 吴宇涵       | 大四   |
+------------+--------------+--------+
2 rows in set (0.016 sec)
```

6. 统计每个学生借书信息，输出每个学生的学号、借书书名和还书日期

```
MariaDB [Library]> SELECT BW.Student_no, B.Book_name, BW.Return_date
    -> FROM Borrow BW
    -> JOIN Book B ON BW.Book_no = B.Book_no;
+------------+--------------------------+-------------+
| Student_no | Book_name                | Return_date |
+------------+--------------------------+-------------+
| K001       | Java 程序设计            | 2024-03-01  |
| K001       | 编译原理                 | 2023-10-09  |
| K002       | Java 程序设计            | 2023-11-03  |
| K002       | 数据库原理及应用         | 2023-10-09  |
| K002       | Java 高级编程            | 2024-04-05  |
| K003       | C++程序设计              | 2024-01-04  |
| K004       | 数据库原理及应用         | 2024-02-05  |
+------------+--------------------------+-------------+
7 rows in set (0.007 sec)
```

7. 查询所有借阅已过期图书的信息，输出学生学号、姓名、书名和还书日期，并按还书日期降序排列

```
MariaDB [Library]> SELECT S.Student_no, S.Student_name, B.Book_name, BW.Return_date
    -> FROM Student S
    -> JOIN Borrow BW ON S.Student_no = BW.Student_no
    -> JOIN Book B ON BW.Book_no = B.Book_no
    -> WHERE BW.Return_date < '2024-04-20'
    -> ORDER BY BW.Return_date DESC;
+------------+--------------+--------------------------+-------------+
| Student_no | Student_name | Book_name                | Return_date |
+------------+--------------+--------------------------+-------------+
| K002       | 王三优       | Java 高级编程            | 2024-04-05  |
| K001       | 刘孔阴       | Java 程序设计            | 2024-03-01  |
| K004       | 吴宇涵       | 数据库原理及应用         | 2024-02-05  |
| K003       | 徐晨皓       | C++程序设计              | 2024-01-04  |
| K002       | 王三优       | Java 程序设计            | 2023-11-03  |
| K001       | 刘孔阴       | 编译原理                 | 2023-10-09  |
| K002       | 王三优       | 数据库原理及应用         | 2023-10-09  |
+------------+--------------+--------------------------+-------------+
7 rows in set (0.018 sec)
```

8. 查询没有借阅过书的学生信息，输出学生姓名和学号

```
MariaDB [Library]> SELECT Student_name, Student_no
    -> FROM Student
    -> WHERE Student_no NOT IN (SELECT Student_no FROM Borrow);
+--------------+------------+
| Student_name | Student_no |
+--------------+------------+
| 张鑫翁       | K005       |
+--------------+------------+
1 row in set (0.017 sec)
```

9. 查询借了"Java 程序设计"但没有借"数据库原理及应用"的读者信息，输出这些学生的学号，并按照学号升序排列

```
MariaDB [Library]> SELECT Student_no
    -> FROM Borrow
    -> JOIN Book ON Borrow.Book_no = Book.Book_no
    -> WHERE Book_name = 'Java 程序设计'
    -> AND Student_no NOT IN (
    ->     SELECT Student_no
    ->     FROM Borrow
    ->     JOIN Book ON Borrow.Book_no = Book.Book_no
    ->     WHERE Book_name = '数据库原理及应用'
    -> )
    -> ORDER BY Student_no ASC;
+------------+
| Student_no |
+------------+
| K001       |
+------------+
1 row in set (0.011 sec)
```

10.插入字段

```
MariaDB [Library]> ALTER TABLE Borrow ADD Borrow_state VARCHAR(20);
Query OK, 0 rows affected (0.035 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
MariaDB [Library]> desc Borrow;
+--------------+------------+------+-----+---------+-------+
| Field        | Type       | Null | Key | Default | Extra |
+--------------+------------+------+-----+---------+-------+
| Student_no   | varchar(20) | NO   | PRI | NULL    |       |
| Book_no      | varchar(20) | NO   | PRI | NULL    |       |
| Return_date  | date       | YES  |     | NULL    |       |
| Borrow_state | varchar(20) | YES  |     | NULL    |       |
+--------------+------------+------+-----+---------+-------+
4 rows in set (0.004 sec)
```

11.创建状态并调用

```
MariaDB [Library]> delimiter $$
MariaDB [Library]> create procedure update_bstate
    -> (in check_date DATE)
    -> begin
    -> update Borrow
    -> set Borrow_state = if(Return_date < check_date,'True','False');
    -> end $$
Query OK, 0 rows affected (0.026 sec)
```

```
MariaDB [Library]> call update_bstate('2024-04-20');
Query OK, 7 rows affected (0.020 sec)

MariaDB [Library]> select * from Borrow;
+------------+---------+-------------+--------------+
| Student_no | Book_no | Return_date | Borrow_state |
+------------+---------+-------------+--------------+
| K001       | T1001   | 2024-03-01  | True         |
| K001       | T1006   | 2023-10-09  | True         |
| K002       | T1001   | 2023-11-03  | True         |
| K002       | T1002   | 2023-10-09  | True         |
| K002       | T1003   | 2024-04-05  | True         |
| K003       | T1005   | 2024-01-04  | True         |
| K004       | T1002   | 2024-02-05  | True         |
+------------+---------+-------------+--------------+
7 rows in set (0.010 sec)
```

12. 修改图书表，在 Book_name 列上增加唯一性索引 Book_name_index，并按 Book_name 降序排列

```
MariaDB [Library]> ALTER TABLE Book ADD UNIQUE INDEX Book_name_index (Book_name DESC);
Query OK, 0 rows affected (0.025 sec)
Records: 0  Duplicates: 0  Warnings: 0

MariaDB [Library]> select * from Book;
+---------+--------------------------+-----------+-------+
| Book_no | Book_name                | Author    | Price |
+---------+--------------------------+-----------+-------+
| T1001   | Java 程序设计            | 李新珊    | 89.50 |
| T1002   | 数据库原理及应用         | 王敏      | 39.00 |
| T1003   | Java 高级编程            | 陈明海    | 63.50 |
| T1004   | 专业英语                 | 张倪宁    | 23.10 |
| T1005   | C++程序设计              | 马天麒    | 83.20 |
| T1006   | 编译原理                 | 王鑫单    | 65.00 |
+---------+--------------------------+-----------+-------+
6 rows in set (0.006 sec)

MariaDB [Library]> SHOW INDEX FROM Book;
+-------+----------+----------------+--------------+------------+-----------+-------------+----------+--------+------+------------+---------+---
----------+---------+
| Table | Non_unique | Key_name        | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Ind
ex_comment | Ignored |
+-------+----------+----------------+--------------+------------+-----------+-------------+----------+--------+------+------------+---------+---
----------+---------+
| Book  |          0 | PRIMARY         |            1 | Book_no     | A         |           6 |     NULL | NULL   |      | BTREE      |         |
          | NO      |
| Book  |          0 | Book_name_index |            1 | Book_name   | D         |           6 |     NULL | NULL   |      | BTREE      |         |
          | NO      |
+-------+----------+----------------+--------------+------------+-----------+-------------+----------+--------+------+------------+---------+---
----------+---------+
2 rows in set (0.012 sec)
```

出现的问题：

建立了索引之后不知道怎么查看是否建立成功

解决方案：

查询了一下，了解到可以使用 SHOW INDEX FROM Book;语句查看索引